

Week 9
Control (Part 2)

ENN581 Robot Motion, Control and Planning

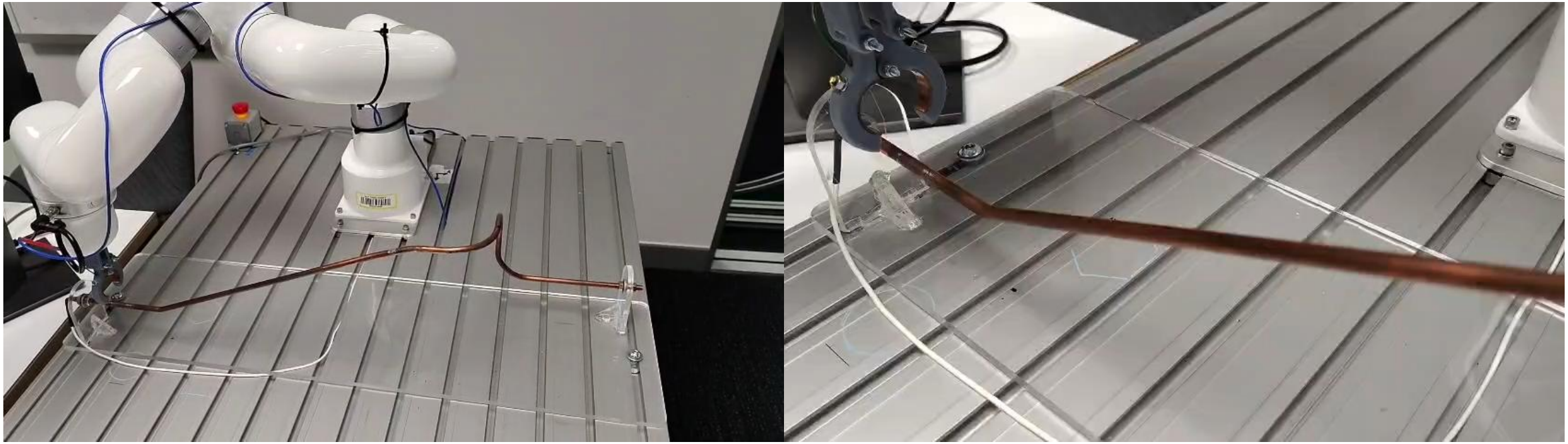
Dr. Krishna Manaswi Digumarti

School of Electrical Engineering and Robotics

ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.



Keep an eye out for scores and feedback this evening

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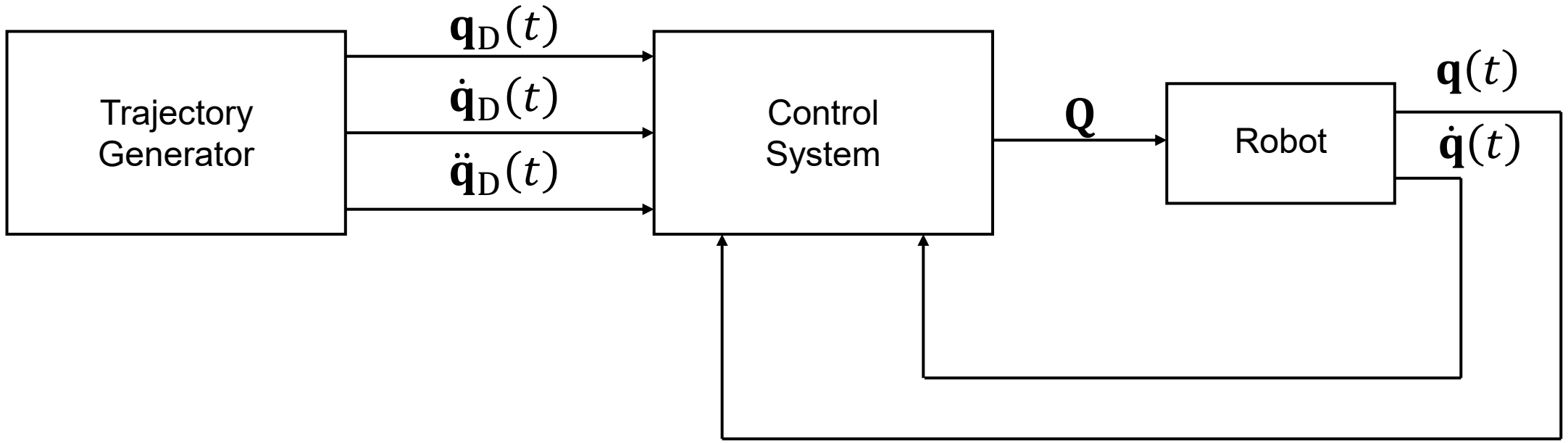
Final Exam

- On 12 November 2026
- Starts at 8:30 am
- GP – P 421
- 2 hrs + 10 min

- Don’t trust this. Check your exam schedules.

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Last week, we learnt



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Today's lecture

- Dealing with non-linear systems
- Combined control law for a manipulator with n joints
- Force control
- Force and position control

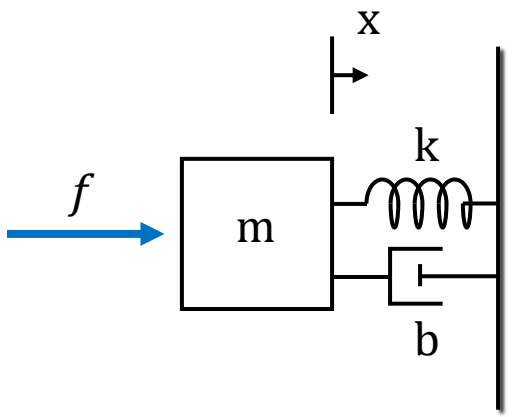
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Simple non-linear systems

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Consider the example of the spring-mass-damper system

net force $= m\ddot{x}$



The equation of motion of this system can be written as

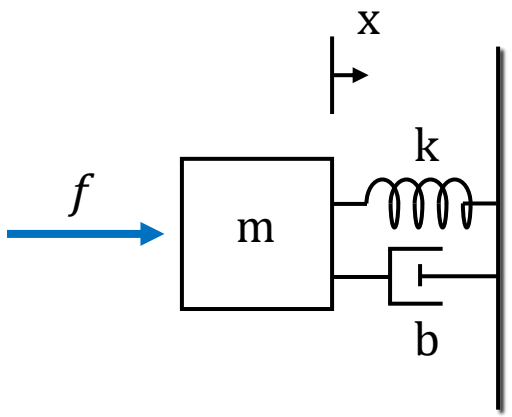
$$m\ddot{x} + b\dot{x} + kx = f$$

This is a second-order, linear differential equation with constant coefficients.

Previously, we looked at various control laws (equations for f) that let us force the system to behave in a desired manner (like being critically damped)

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Let us introduce a non-linear spring



The spring force in this case is proportional to the cube of position

$$f_{spring} = \mathrm{kx}^3$$

The equation of motion of this system can be written as

$$\mathrm{m}\ddot{\mathrm{x}} + \mathrm{b}\dot{\mathrm{x}} + \mathrm{kx}^3 = f$$

$\alpha f + \beta$

This is a second-order, non-linear differential equation with constant coefficients.

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We can define a partitioned control law as before

$$f = \alpha \acute{f} + \beta$$

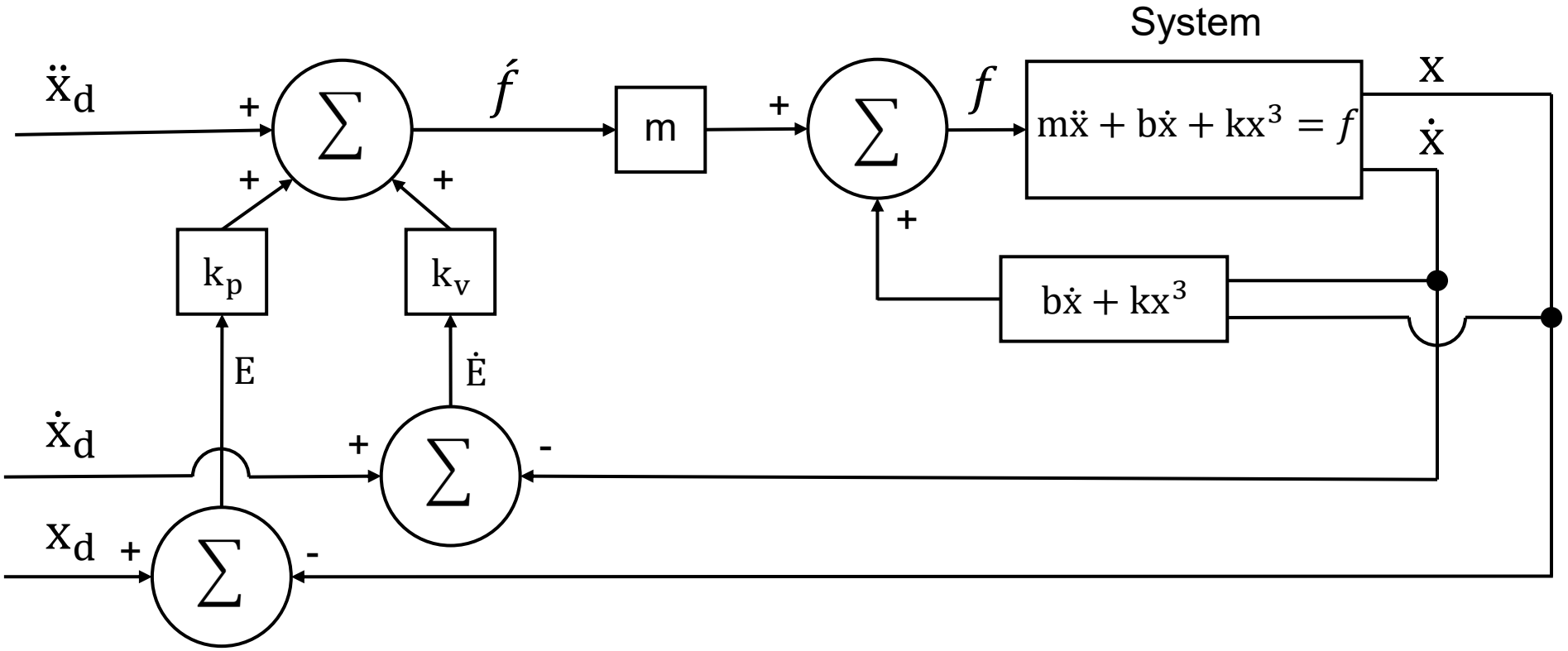
We choose $\alpha = \mathrm{m}$ and $\beta = \mathrm{b}\dot{\mathrm{x}} + \mathrm{k}\mathrm{x}^3$

Servo portion is still the same as before

$$\acute{f} = \ddot{\mathrm{x}}_\mathrm{d} + \mathrm{k}_\mathrm{v}\dot{\mathrm{E}} + \mathrm{k}_\mathrm{p}\mathrm{E} \qquad E = \mathrm{x}_\mathrm{d} - \mathrm{x}$$

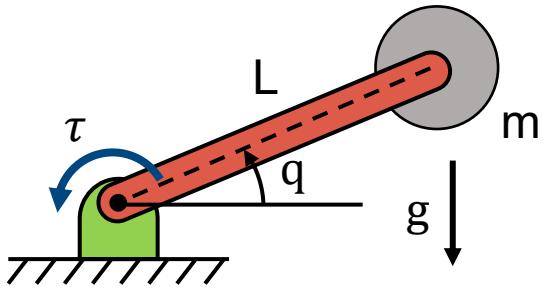
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The block diagram of the non-linear control system



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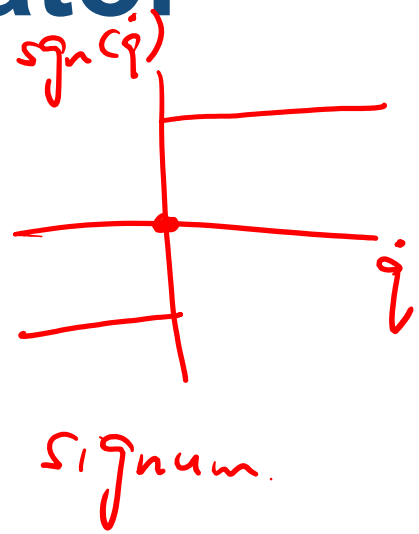
Example 2: A single link manipulator



$q, \dot{q}, \ddot{q}$

Assume two types of friction

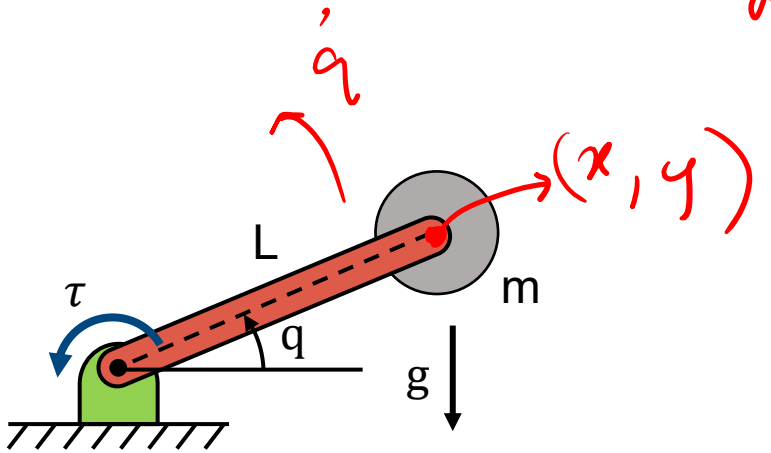
1. Viscous friction: $\tau_v = \mathrm{b}_v \dot{\mathrm{q}}$
2. Coulomb friction: $\tau_c = \mathrm{b}_c sgn(\dot{\mathrm{q}})$



Step 1: write down the equation of motion

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Example 2: A single link manipulator



Assume two types of friction

1. Viscous friction: $\tau_v = \mathrm{b}_v \dot{\mathrm{q}}$
2. Coulomb friction: $\tau_c = \mathrm{b}_c sgn(\dot{\mathrm{q}})$

$$\tau = \mathrm{mL}^2\ddot{\mathrm{q}} + \mathrm{b}_v\dot{\mathrm{q}} + \mathrm{b}_c sgn(\dot{\mathrm{q}}) + \mathrm{mgL}\cos(\mathrm{q})$$

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A suitable control law for the single link system

Dynamics equation of motion

$$\tau = \mathrm{mL}^2\ddot{\mathrm{q}} + \mathrm{b}_v\dot{\mathrm{q}} + \mathrm{b}_c sgn(\dot{\mathrm{q}}) + \mathrm{mgL}\cos(\mathrm{q})$$

Split the control law

$$\tau = \alpha\acute{\tau} + \beta$$

We choose $\alpha = \mathrm{mL}^2$

and $\beta = \mathrm{b}_v\dot{\mathrm{q}} + \mathrm{b}_c sgn(\dot{\mathrm{q}}) + \mathrm{mgL}\cos(\mathrm{q})$

Servo portion is the same as before

$$\acute{\tau} = \ddot{\mathrm{q}}_\mathrm{d} + \mathrm{k}_\mathrm{v}\dot{\mathrm{E}} + \mathrm{k}_\mathrm{p}\mathrm{E}$$

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General procedure in simple cases

- First, we determined the equation of motion for the system
- Then, we computed a nonlinear model-based portion of the control law that cancels the non-linearities in the system
- We then reduced the system to a linear one which can be controlled by the linear servo law

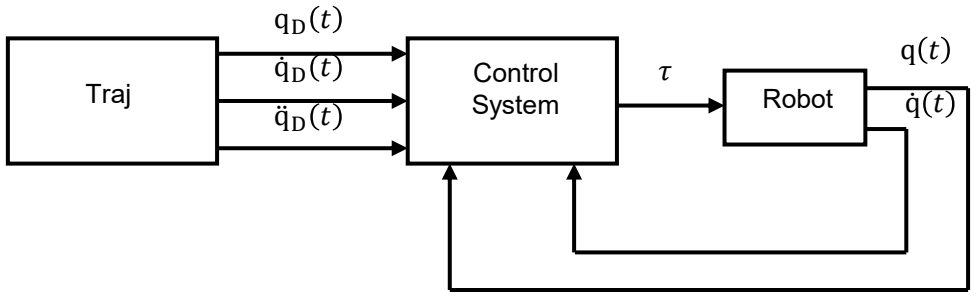
To follow this process, we need to know the parameters of the system (k, b_c, etc.) and its structure (cubic spring force, Coulomb type friction, etc.).
In practice, these are not always known.

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Multi-input, multi-output (MIMO) systems

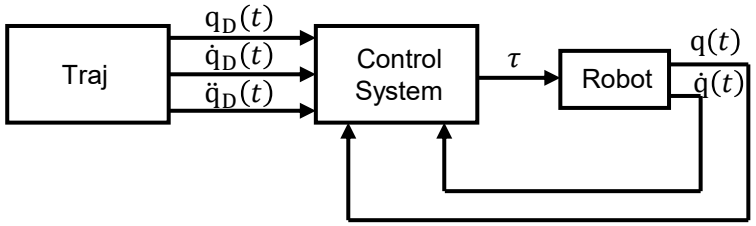
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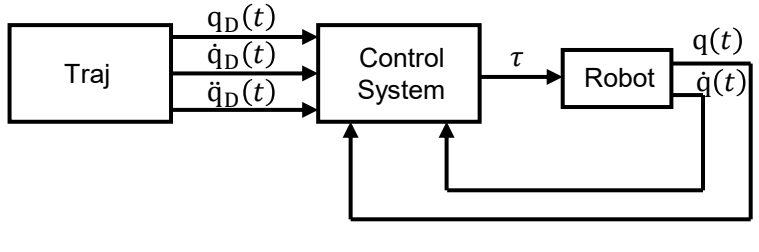
Recall: Independent joint control

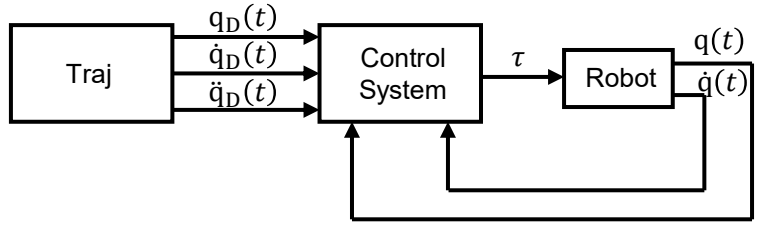


In general, for a robot with n joints, each line represents an $n \times 1$ vector.

We simplifed this into n independent control problems where each line is a single entity.







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Control law partitioning in MIMO systems

- We have a vector of joint positions, velocities and accelerations.
- The control law must compute a vector of joint actuations.
- The model-based portion is

$$\underset{n\times 1}{F} = \underset{n\times n}{\alpha}\underset{n\times 1}{\acute{F}} + \underset{n\times 1}{\beta}$$

- α is chosen such that it decouples the n equations.
- The servo portion is

$$\underset{n\times 1}{\acute{F}} = \underset{n\times 1}{\ddot{\mathrm{X}}_{\mathrm{d}}} + \underset{n\times n}{\mathrm{K_v}}\underset{n\times 1}{\dot{\mathrm{E}}} + \underset{n\times n}{\mathrm{K_p}}\underset{n\times 1}{\mathrm{E}}$$

Note:

- α is not a diagonal matrix
- $\mathrm{K_v}$ and $\mathrm{K_p}$ are usually diagonal matrices

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Recall: dynamics of a manipulator

We represented the dynamics of a manipulator as

$$\underset{n\times 1}{\mathbf{Q}} = \underset{n\times n}{\mathbf{M}(\mathbf{q})}\underset{n\times 1}{\ddot{\mathbf{q}}} + \underset{n\times 1}{\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})} + \underset{n\times 1}{\boldsymbol{g}(\mathbf{q})} + \underset{n\times 1}{\boldsymbol{f}(\dot{\mathbf{q}})}$$

Such a system can be controlled by a partitioned control law as follows

$$\mathbf{Q} = \boldsymbol{\alpha}\acute{\mathbf{Q}} + \boldsymbol{\beta}$$

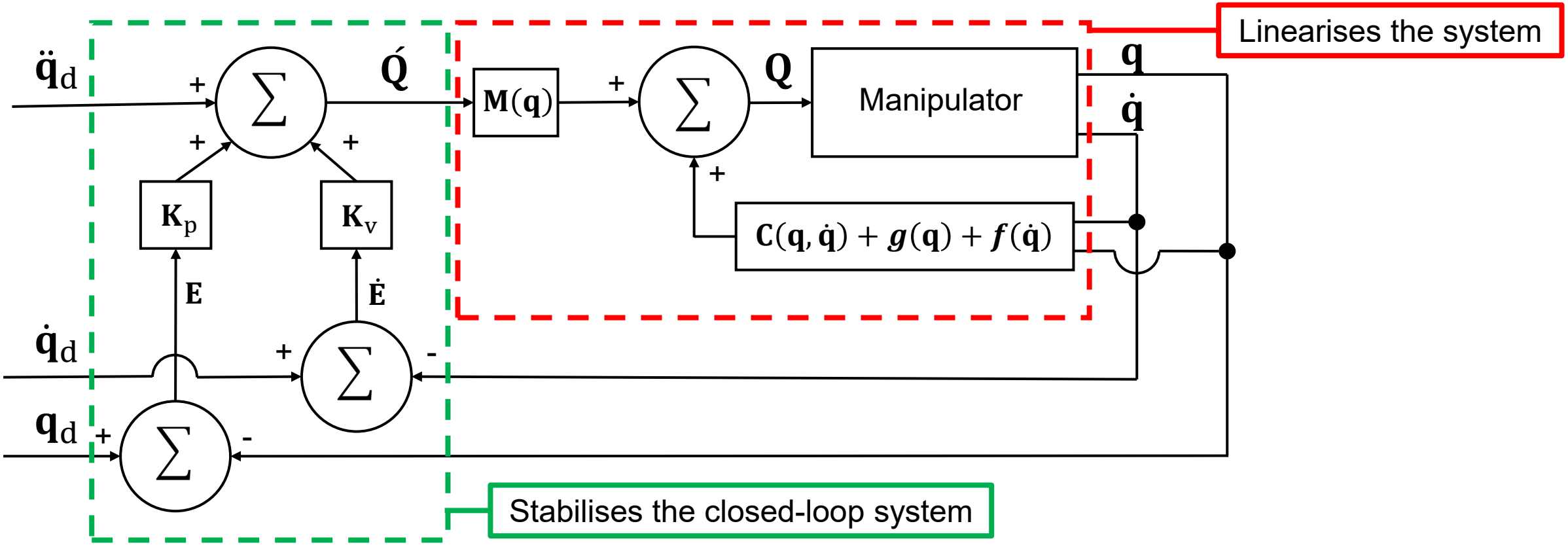
We choose $\boldsymbol{\alpha} = \mathbf{M}(\mathbf{q})$ and $\boldsymbol{\beta} = \mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) + \boldsymbol{g}(\mathbf{q}) + \boldsymbol{f}(\dot{\mathbf{q}})$

With the servo law

$$\underset{n\times 1}{\acute{\mathbf{Q}}} = \underset{n\times 1}{\ddot{\mathbf{q}}_{\mathrm{d}}} + \underset{n\times n}{\mathbf{K}_{\mathrm{v}}}\underset{n\times 1}{\dot{\mathbf{E}}} + \underset{n\times n}{\mathbf{K}_{\mathrm{p}}}\underset{n\times 1}{\mathbf{E}}$$

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Block diagram of the MIMO control system of a manipulator



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Closed loop control system is decoupled

- The closed loop control system is characterised by the equation

$$\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = 0$$

- Since $\mathbf{K}_P$ and $\mathbf{K}_v$ are diagonal matrices, the above is a linear and decoupled system of the form

$$\ddot{e} + k_{vi}\dot{e} + k_{pi}e = 0$$

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The closed loop dynamics given by $\ddot{\mathrm{E}} + \mathrm{K_v}\dot{\mathrm{E}} + \mathrm{K_p}\mathrm{E} = 0$ is an ideal case

In practice, there are two main issues:

1. The manipulator model may be inaccurate i.e.,
   the matrices $\mathbf{M}(\mathbf{q})$, $\mathbf{C}(\mathrm{q}, \dot{\mathrm{q}})$, $\boldsymbol{g}(\mathrm{q})$ and $\boldsymbol{f}(\dot{\mathrm{q}})$ may not be known precisely.
2. Discretization of the control law when implementing it on a computer.

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What is the problem with discretization?

- The controller, when implemented on a computer, updates at discrete intervals of time.
- Sensors are read at discrete points in time (based on sampling rate) and the actuator commands are updated.
- This does not happen continuously.
- Computations take a finite amount of time and restrict the rate at which joint torques can be updated.

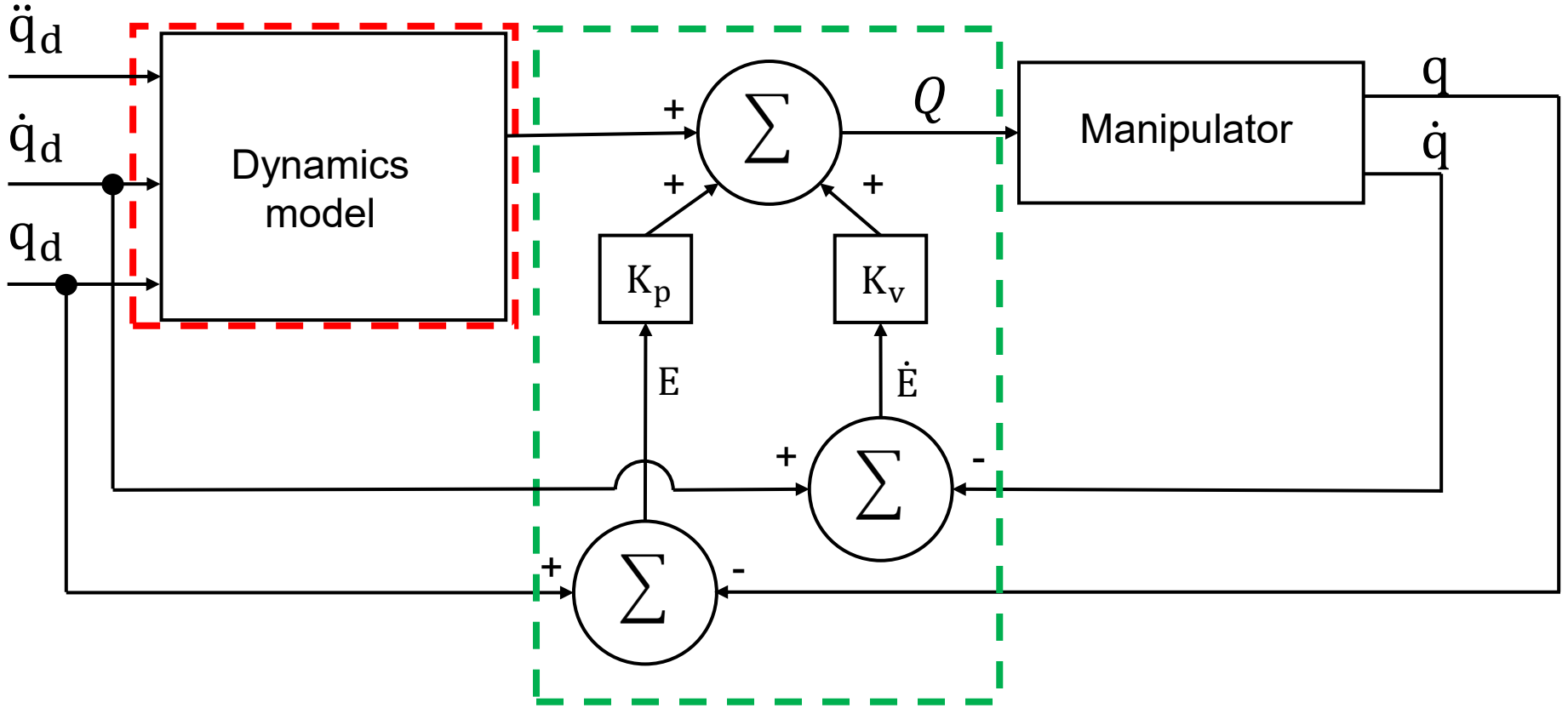
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Which computations take time?

- The dynamics computed in the model-based part take time (recall the lecture on dynamics)
- The servo part is quicker to update as only computation of error and multiplication with gains is happening in this block

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One approach is to move the model-based part to the outer loop



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Why is the manipulator model imprecise?

- Friction is hard to model
- Robot wears down
- Objects being grasped can change the weight distribution

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How do we deal with an imprecise model?

System dynamics are given by $\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{g}(\mathbf{q}) + \boldsymbol{f}(\dot{\mathbf{q}})$

Since we don't know the model parameters exactly, the control law is also imprecise

$$\mathbf{Q} = \alpha\acute{\mathbf{Q}} + \beta \qquad \alpha = \widehat{\mathbf{M}}(\mathrm{q}) \qquad \beta = \hat{\mathbf{C}}(\mathrm{q}, \dot{\mathrm{q}}) + \hat{\boldsymbol{g}}(\mathrm{q}) + \hat{\boldsymbol{f}}(\mathrm{q}, \dot{\mathrm{q}})$$

Hat indicates our imprecise knowledge of model parameters

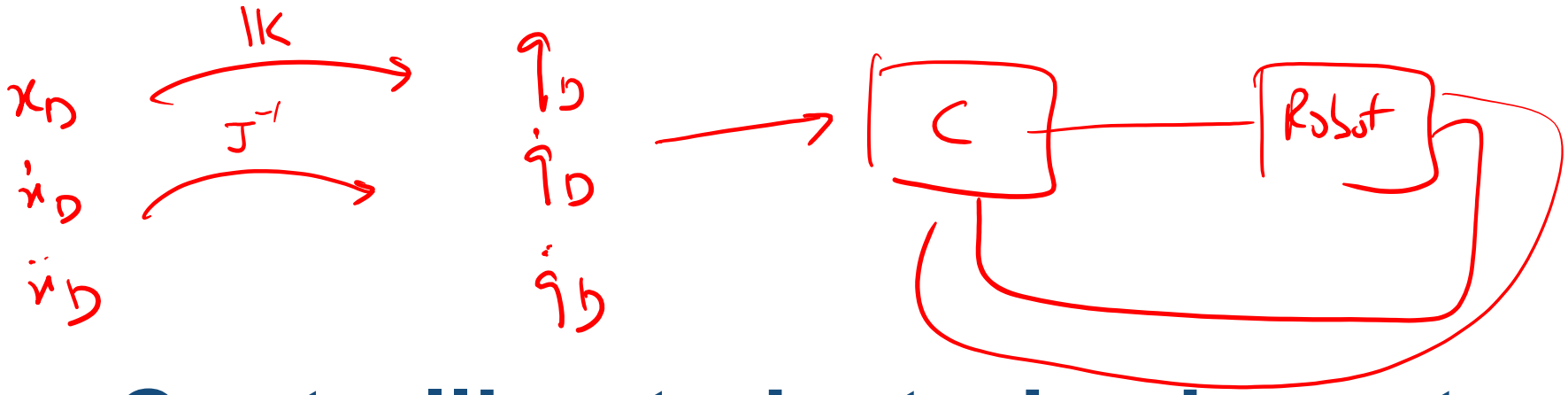
$$\acute{\mathbf{Q}} = \ddot{\mathrm{q}}_{\mathrm{d}} + \mathrm{K}_{\mathrm{v}}\dot{\mathrm{E}} + \mathrm{K}_{\mathrm{p}}\mathrm{E}$$

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Closed-loop system dynamics in case of an imprecise model

$$\ddot{\mathrm{E}} + \mathrm{K_v}\dot{\mathrm{E}} + \mathrm{K_p}\mathrm{E} = \widehat{\mathbf{M}}^{-1}(\mathrm{q})\left[\left(\mathbf{M}(\mathrm{q}) - \widehat{\mathbf{M}}(\mathrm{q})\right)\ddot{\mathrm{q}} + \left(\mathbf{C}(\mathrm{q},\dot{\mathrm{q}}) - \hat{\mathbf{C}}(\mathrm{q},\dot{\mathrm{q}})\right) + \left(\boldsymbol{g}(\mathrm{q}) - \hat{\boldsymbol{g}}(\mathrm{q})\right) + \left(\boldsymbol{f}(\mathrm{q},\dot{\mathrm{q}}) - \hat{\boldsymbol{f}}(\mathrm{q},\dot{\mathrm{q}})\right)\right]$$

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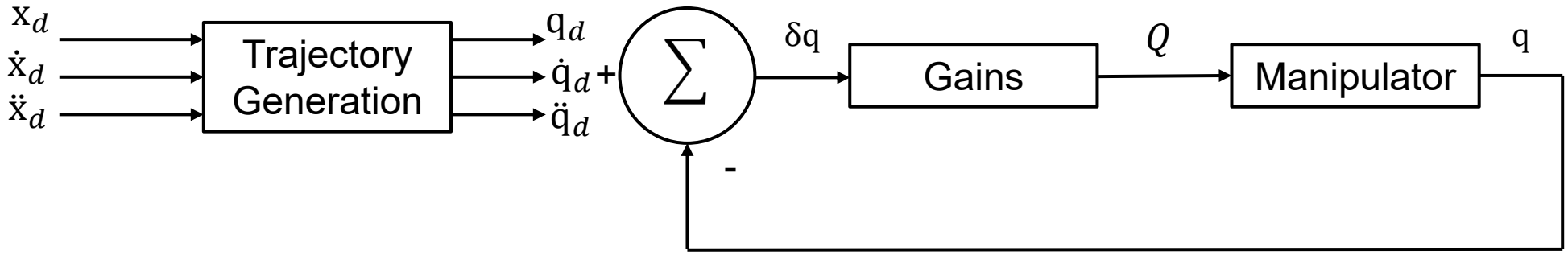


Controlling trajectories in cartesian space

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We may want to implement the control in cartesian space

- We know how to obtain a time series of points in joint space when we desire a path in cartesian space.
- We could use a setup like the one below where a trajectory generator converts the cartesian space coordinates to joint space coordinates.



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Is the trajectory generation part efficient?

What are all the computations that we need to perform?

$$\mathrm{q} = \mathcal{K}^{-1}(\mathrm{x})$$

$$\dot{\mathrm{q}} = J^{-1}(\mathrm{q})\dot{\mathrm{x}}$$

$$\ddot{\mathrm{q}} = \dot{J}^{-1}(\mathrm{q})\dot{\mathrm{x}} + J^{-1}(\mathrm{q})\,\ddot{\mathrm{x}}$$

All these are computationally intensive. Is there an alternative?

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Repose the problem in the cartesian space

Equation of motion of the manipulator in cartesian space is

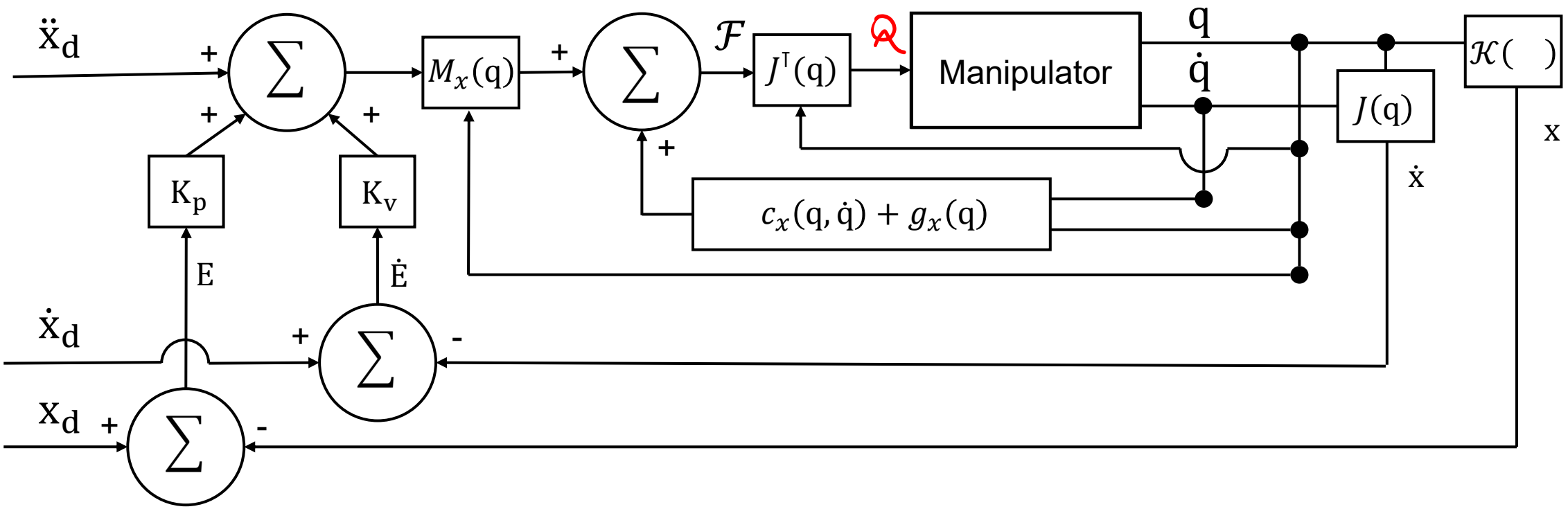
$$\boldsymbol{\mathcal{F}} = \mathbf{M}_x(\mathbf{q})\ddot{\mathbf{x}} + \mathbf{C}_x(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{g}_x(\mathbf{q})$$

Solve for the control force $\mathcal{F}$, given a desired trajectory. Then compute joint torques.

$$\mathbf{Q} = \mathbf{J}^{\mathsf{T}}(\mathbf{q})\boldsymbol{\mathcal{F}}$$

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Block diagram of a Cartesian-based controller



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Force control

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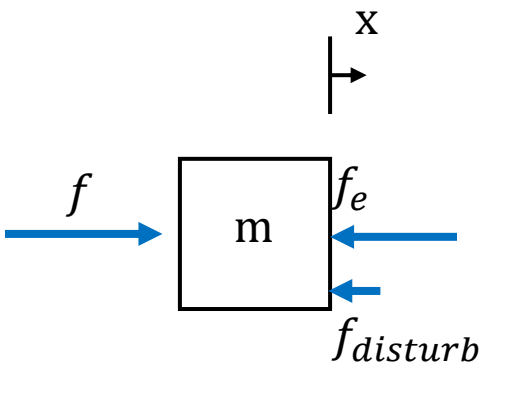
What is the goal of force control?

- To make the end effector apply a desired force on the environment.
- How do we control the joint torques to be able to do this?

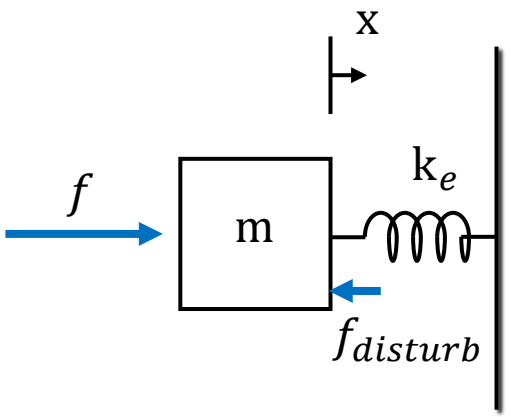
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Example: Spring mass system

$f - f_e - f_{dist} = m\ddot{x}$



Goal:
determine an appropriate control force $f(t)$,
that results in a desired interaction force f_e
between the mass and the environment (wall).



$f_e = k_e x$

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The equation describing this system is

$$f = \mathrm{m}\ddot{\mathrm{x}} + f_{disturb} + f_e$$

$f_e = k x$

$$f = \mathrm{m}\mathrm{k}_e^{-1}\ddot{f}_e + f_{disturb} + f_e$$

$$f_e = \mathrm{k}_e \mathrm{x} \Leftrightarrow \mathrm{x} = \mathrm{k}_e^{-1} f_e$$

We can write a partitioned controller as before

$$f = \alpha \acute{f} + \beta$$

we choose $\alpha = \mathrm{m}\mathrm{k}_e^{-1}$ and $\beta = f_{disturb} + f_e$

We then arrive at a control law

$$f = \mathrm{m}\mathrm{k}_e^{-1}\left[\ddot{f}_\mathrm{d} + \mathrm{k}_\mathrm{v}\dot{\mathrm{E}} + \mathrm{k}_\mathrm{p}\mathrm{E}\right] + f_{disturb} + f_e$$

f_d = desired interaction force

$$\mathrm{E} = f_d - f_e$$

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Closed loop control equation is

$$\ddot{\mathrm{E}} + \mathrm{K_v}\dot{\mathrm{E}} + \mathrm{K_p}\mathrm{E} = 0$$

What is the problem with our approach?

- The control law has $f_{disturb}$ in it.
- We don't have sufficient knowledge of this term.
- Therefore, our control law is infeasible.

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What can we do about $f_{disturb}$?

- Option 1: ignore it in the control law.

$$f = \mathrm{m}k_e^{-1}\left[\ddot{f}_\mathrm{d} + \mathrm{k_v}\dot{\mathrm{E}} + \mathrm{k_p}\mathrm{E}\right] + f_e = \mathrm{m}k_e^{-1}\ddot{f}_e + f_{disturb} + f_e \quad \Rightarrow \ddot{\mathrm{E}} + \mathrm{k_v}\dot{\mathrm{E}} + \mathrm{k_p}\mathrm{E} = f_{disturb}/\mathrm{m}k_e^{-1}$$

At steady state, the error is $\mathrm{E} = \dfrac{f_{disturb}}{\mathrm{m}k_e^{-1}\mathrm{k_p}}$

- Option 2: use f_d in place of $f_{disturb} + f_e$.

$$f = \mathrm{m}k_e^{-1}\left[\ddot{f}_\mathrm{d} + \mathrm{k_v}\dot{\mathrm{E}} + \mathrm{k_p}\mathrm{E}\right] + f_\mathrm{d} = \mathrm{m}k_e^{-1}\ddot{f}_e + f_{disturb} + f_e \Rightarrow \ddot{\mathrm{E}} + \mathrm{K_v}\dot{\mathrm{E}} + \left(1/\mathrm{m}k_e^{-1} + \mathrm{k_p}\right)\mathrm{E} = f_{disturb}/\mathrm{m}k_e^{-1}$$

At steady state, the error is $\mathrm{E} = \dfrac{f_{disturb}}{1 + \mathrm{m}k_e^{-1}\mathrm{k_p}}$

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Which control law is better?

Case 1

$$f = \mathrm{m}k_e^{-1}\left[\ddot{f}_\mathrm{d} + \mathrm{k_v}\dot{\mathrm{E}} + \mathrm{k_p}\mathrm{E}\right] + f_e$$

Steady state error is $\mathrm{E} = \dfrac{f_{disturb}}{\mathrm{m}k_e^{-1}\mathrm{k_p}}$

Case 2

$$f = \mathrm{m}k_e^{-1}\left[\ddot{f}_\mathrm{d} + \mathrm{k_v}\dot{\mathrm{E}} + \mathrm{k_p}\mathrm{E}\right] + f_\mathrm{d}$$

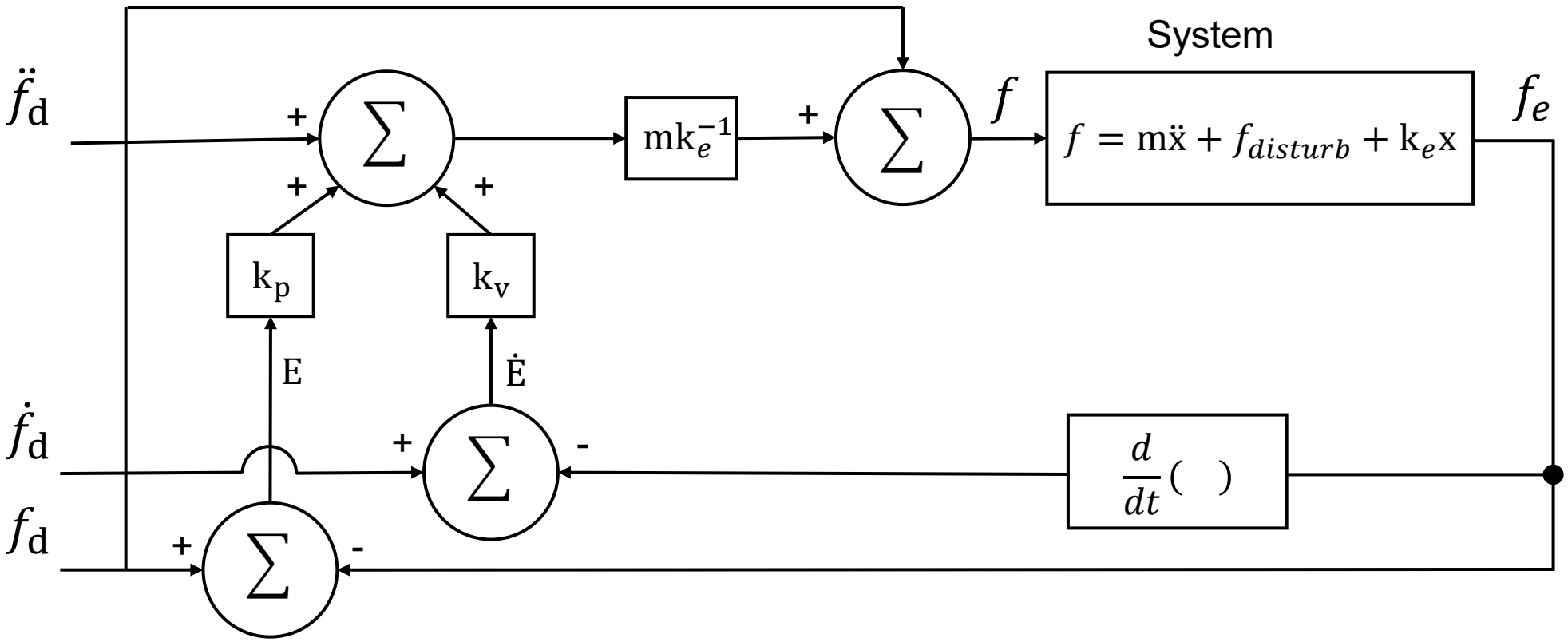
Steady state error is $\mathrm{E} = \dfrac{f_{disturb}}{1 + \mathrm{m}k_e^{-1}\mathrm{k_p}}$

If the environment is stiff, $\mathrm{k_e}$ is large

Steady state error in case 2 is smaller than that in case 1

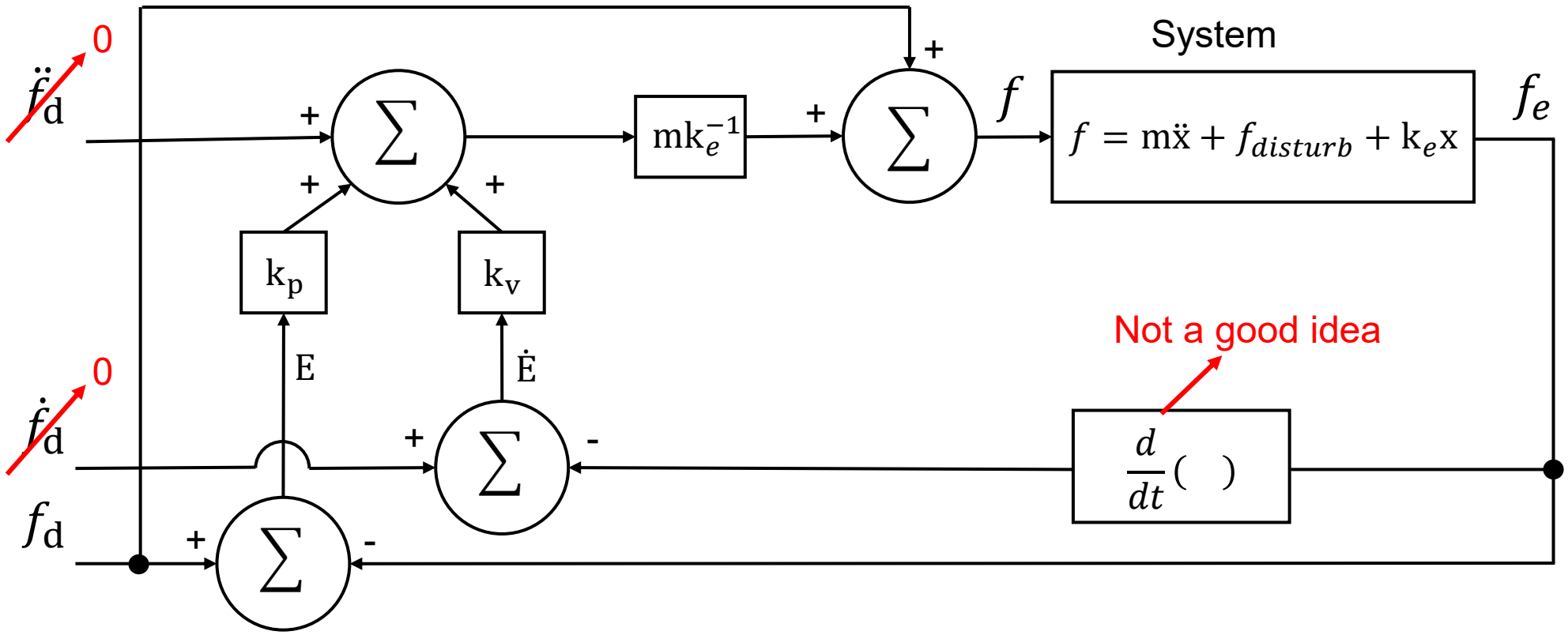
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Block diagram of a force control system



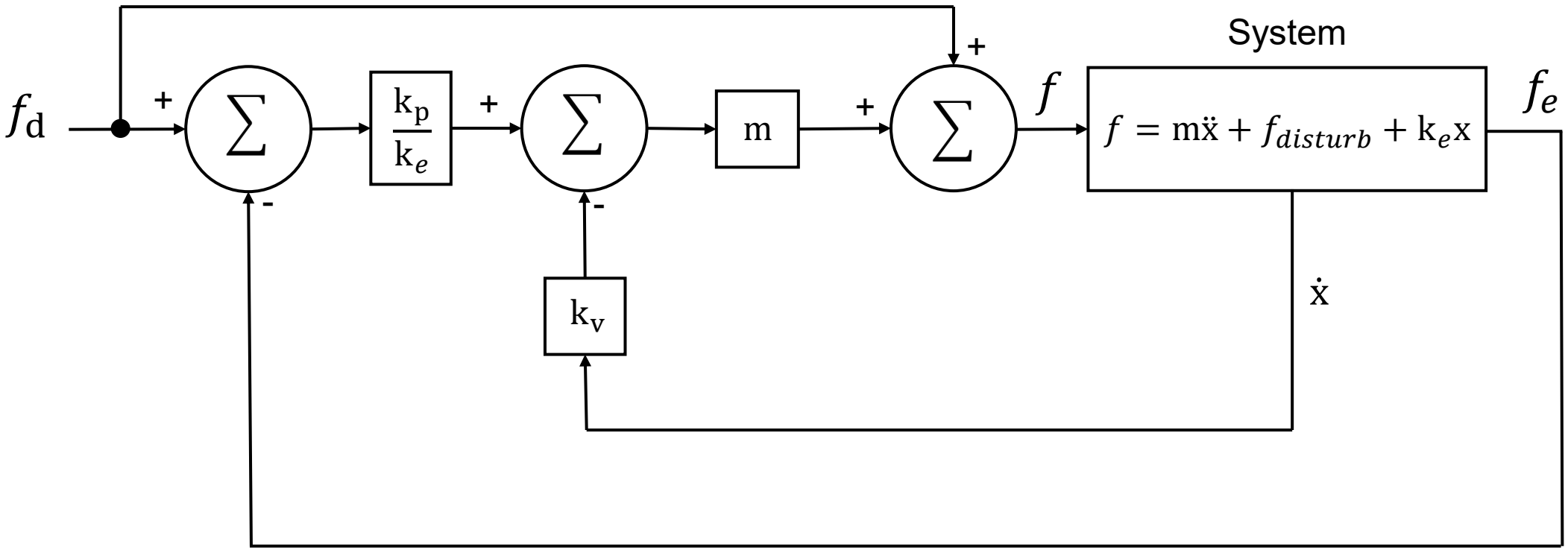
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Block diagram of a force control system



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A more practical block diagram



Control law is $f = m\left[k_e^{-1} k_p E - k_v \dot{x}\right] + f_d$

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In general, we don't know the stiffness k_e

Usually chosen to be a high value.

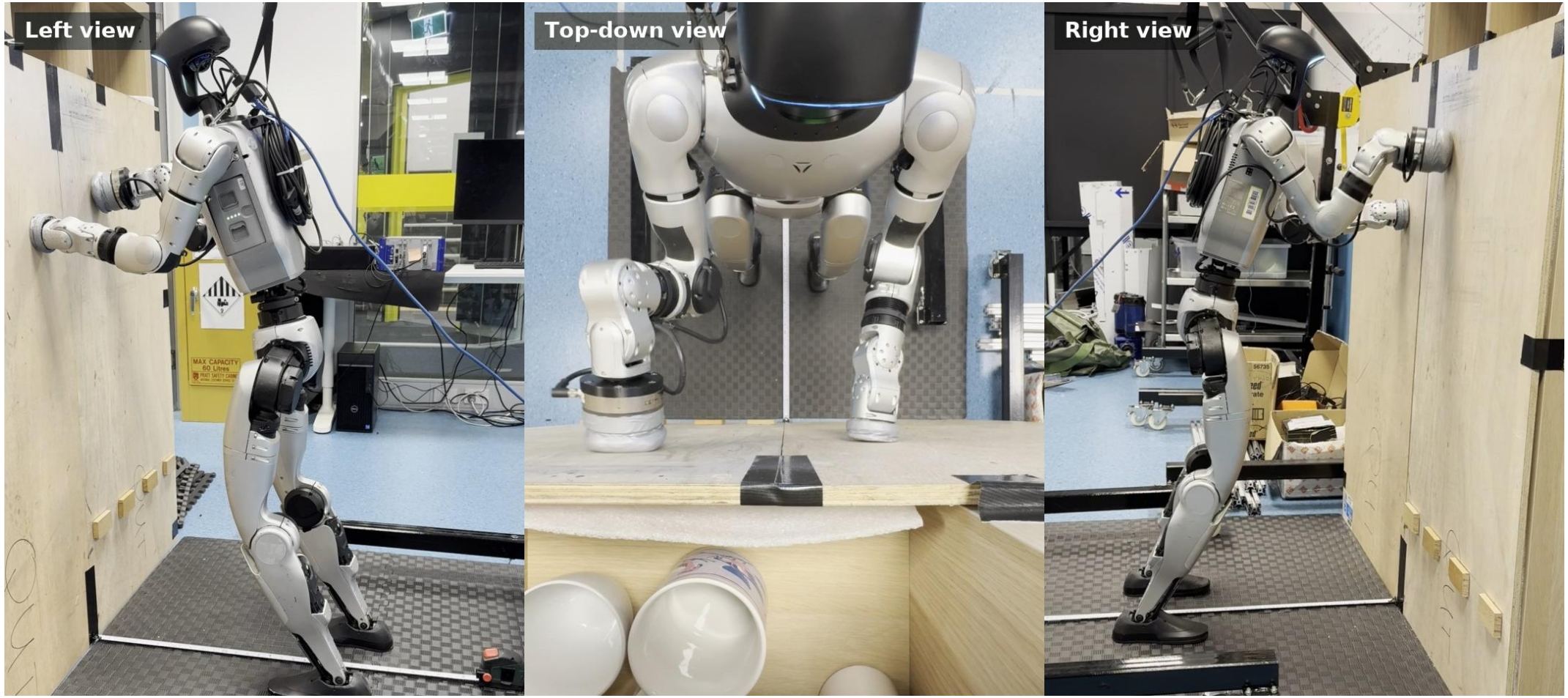
Controller is designed to be robust to small changes in k_e.

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Hybrid Force and Position Control

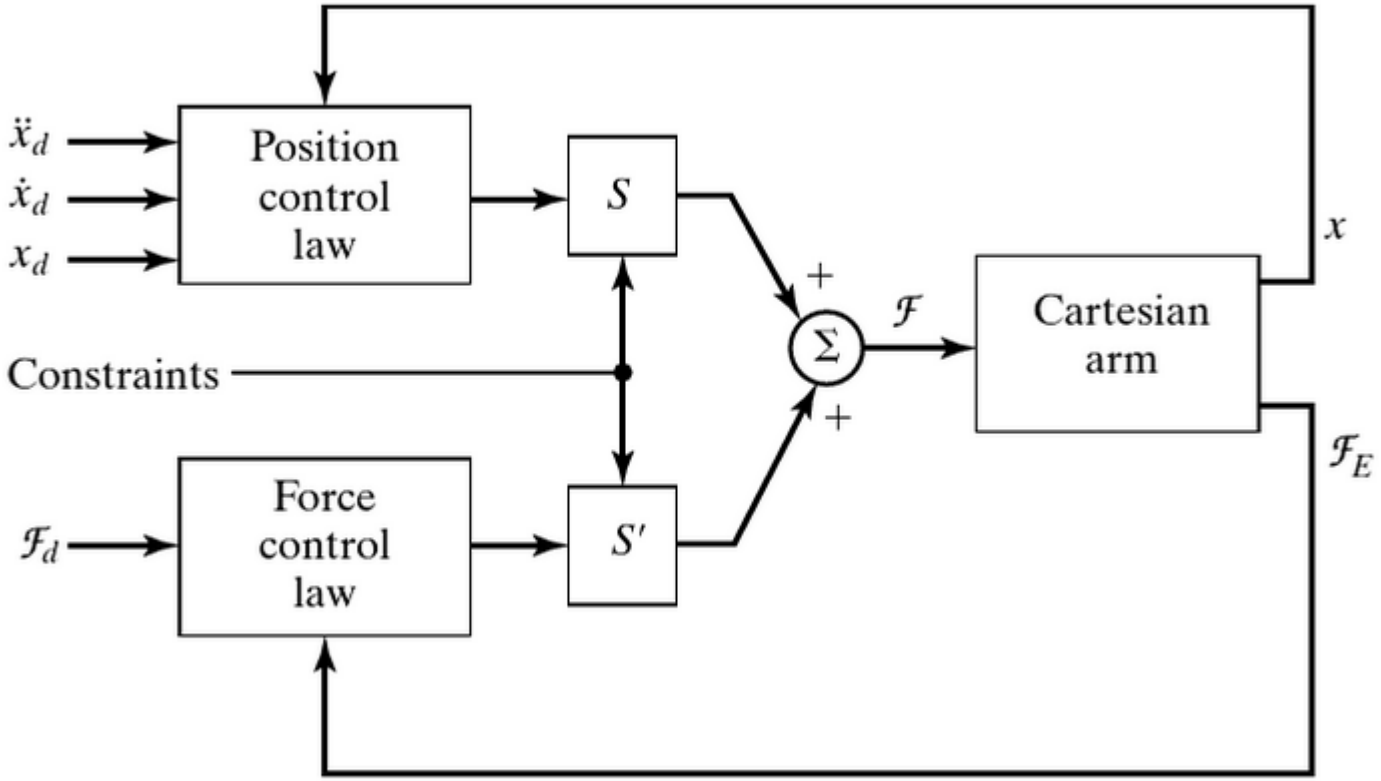
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A more general task requires both position and force control



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A combination of force and position control



S and S' are diagonal matrices with either 0 or 1 on the diagonal.

They are complementary to each other.

They indicate whether a joint is in force control mode or position control mode.

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Further reading

- Chapters 10 and 11 of Introduction to Robotics by Craig
- Chapter 11 of Modern Robotics by Lynch and Park

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Summary

- Computed torque control
  - Cancel out nonlinearity using the model-based part
  - Ensure stability of the system using the servo part
- Multi-input, multi-output systems
- Cartesian-based control
- Force control
- Hybrid position and force control

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Next lecture

- No class on 21st of September
  - Keep an eye on canvas for additional content.
- Next lecture is on 28th of September
- We will switch to mobile robot kinematics
- Assessment 2.4 will be open on 18.09.2026

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